$$\langle X|\,\hat{P}\hat{\rho}_{\mathcal{S}}(t)\,|X'\rangle = -\mathrm{i}\frac{\partial}{\partial X}\,\langle X|\,\hat{\rho}_{\mathcal{S}}(t)\,|X'\rangle \equiv -\mathrm{i}\frac{\partial}{\partial X}\rho_{\mathcal{S}}(X,X',t), \qquad (5.27)$$